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Senior Machine Learning Engineer – Generative AI
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PROFESSIONAL SUMMARY

- Results-oriented **Senior Data Engineer / ML Engineer** with **9+ years** of experience delivering large-scale data solutions, production-grade ML systems, and enterprise-grade analytics platforms across domains like insurance, healthcare, and tech.
- Strong expertise in **building robust data pipelines**, including **ETL/ELT frameworks**, **streaming architectures**, and **feature stores**, using tools like **Airflow**, **Talend**, **Spark**, **Snowflake**, and **DBT**.
- Proven ability to **operationalize machine learning models** in production using **AWS SageMaker**, **Docker**, **Kubernetes (EKS)**, and CI/CD automation, supporting use cases like fraud detection, churn prediction, demand forecasting, and GenAI assistants.
- Deep experience in **MLOps practices**, including **model versioning**, **batch and real-time inference**, **monitoring (Prometheus, Grafana, Splunk)**, and **pipeline reproducibility** across environments.
- Hands-on with **GenAI systems** — built end-to-end **LLM-driven chatbots** using **GPT-4o**, **Claude**, **RAG**, **LangChain**, and evaluated Bedrock-hosted models for summarization, Q&A, and document parsing.
- Designed and deployed **streaming and micro-batch pipelines** using **Kafka**, **Spark Streaming**, and integrated ML outputs into real-time decision support tools.
- Advanced SQL and PL/SQL skills; experienced in **data modeling**, **schema optimization**, and **query performance tuning** across relational and columnar data stores.
- Proficient in building cloud-native architectures on **AWS**, leveraging services like **S3**, **EC2**, **Step Functions**, **EKS**, and implementing IaC using **Terraform**.
- Strong DevOps and CI/CD advocate, having automated end-to-end deployments using **Jenkins**, **Git**, and **GitHub Actions**, with focus on testing, rollback, and release velocity.
- Collaborative and cross-functional — partnered with data scientists, analysts, product managers, and engineers to turn prototypes into production-ready systems, while mentoring junior engineers and contributing to engineering best practices.
- Adept at working in **Agile and Scrum environments**, with strong project ownership, documentation, and delivery across rapidly evolving product and platform teams.
- Familiar with scientific computing and statistical analysis libraries including **SciPy**, **Seaborn**, and **Matplotlib**, with use in model diagnostics and reporting.
- Exposure to **AML-adjacent techniques** such as **fraud detection**, **anomaly detection**, and high-risk transaction scoring using XGBoost and time-series features.

Technical Stack: -

Category	Skills & Tools
Programming Languages	Python, SQL, PL/SQL, Shell Scripting, R, Java
ML & AI	Scikit-learn, XGBoost, TensorFlow, PyTorch, BERT, GPT-4o, LangChain, LLM Evaluation, SageMaker
GenAI & NLP	RAG, FAISS, Prompt Engineering, Claude, Titan, Cohere, OpenAI API, AWS Bedrock, Hugging Face, SpaCy, NLTK
Data Engineering	Apache Spark, PySpark, Azure Databricks, Airflow, DBT, Talend, Pentaho, Feature Stores, ETL/ELT Pipelines

Cloud & Infrastructure	AWS (S3, EC2, EKS, SageMaker, Step Functions), Azure (Databricks), Kubernetes, Docker, Terraform, Jenkins, Git
Databases	Snowflake, Oracle, SQL Developer, PostgreSQL, MongoDB, Elasticsearch, NoSQL & Caching
Data Formats	JSON, Parquet, Avro, XML
Streaming & Messaging	Kafka, Spark Streaming, RabbitMQ
Monitoring & Logging	Splunk, Prometheus, Grafana, Custom Metrics, Log Aggregation
Visualization & BI	Streamlit, Tableau, Power BI, Looker, Matplotlib, Seaborn, Plotly
Orchestration & CI/CD	Airflow, AWS Step Functions, Jenkins, GitHub Actions, CI/CD Pipelines
Tools & Platforms	Jupyter, Azure Databricks, Jira, Confluence, Google Workspace, REST APIs
Dev Methodology	Agile, Scrum, DevOps, MLOps
Operating Systems	Windows, Linux, Unix, Mac OS
Category	Skills & Tools

Education

Degree	Institution	Year	Location
Bachelor of Technology in Computer Science	GITAM University	2014	Visakhapatnam, India

PROFESSIONAL EXPERIENCE

Indeed, Austin, Texas
Senior Machine Learning Engineer – Generative AI

July'23- Present

- Led the development of end-to-end ML and GenAI systems to improve internal analytics workflows, product insights, and user-job relevance across Indeed's ecosystem.
- Designed and deployed a **BERT-based semantic matching system** using **VectorDB (FAISS)** to match user profiles with job listings, boosting search relevance and application conversion.
- Built **causal inference pipelines** to measure the impact of employer interventions (e.g., pricing and visibility), directly influencing product strategy and revenue optimization.
- Leveraged Azure Databricks for distributed data preparation and model training, enhancing pipeline scalability and reducing runtime by 30%.
- Developed an **Internal GenAI-powered chatbot** to enable natural language access to product KPIs and marketplace metrics for Indeed’s **Senior Leadership Team (SLT)**, accelerating data-driven decision-making and reducing dependency on manual analyst workflows.

- **Integrated AWS Bedrock-hosted models (Claude, Titan)** with a **LangChain-based Retrieval-Augmented Generation (RAG) pipeline**, combining real-time embeddings (FAISS) and Snowflake queries to generate grounded, context-aware responses.
- Built an intuitive **Streamlit interface** tailored for non-technical users, featuring prompt chaining, metadata-aware citation generation, and dynamic query resolution — enabling leadership to ask complex, cross-domain product questions and receive explainable insights in seconds.
- Created prompt chains and embedding-based retrieval systems to ensure accurate, grounded, and interpretable responses across diverse product domains.
- **Evaluated and integrated AWS Bedrock models** (Claude, Titan, Cohere) by conducting **side-by-side benchmarking** for various GenAI tasks such as summarization, product data Q&A, and document parsing; provided model selection guidance based on **latency, accuracy, and cost trade-offs** for internal adoption.
- Leveraged **Amazon SageMaker** to run **scalable batch inference pipelines** and perform **offline prompt tuning** on large volumes of historical product data, supporting use cases like auto-generated summaries, insight surfacing, and grounding validation during GenAI system bootstrapping.
- Implemented AWS Step functions to automate and orchestrate the **Amazon SageMaker** related tasks such as publishing data into S3, training ML model and deploying it for prediction.
- Built reusable **prompt templates** and **LLM evaluation frameworks**, reducing turnaround time for GenAI feature development and experimentation.
- Collaborated cross-functionally with engineers, analysts, and scientists to embed GenAI into dashboards and workflows, driving broader org-wide adoption.
- Partnered with UX and product stakeholders to improve GenAI assistant usability, increasing task success rates and reducing response latency for internal users.
- Authored internal documentation and architectural guides to scale LLM-based tools across teams and streamline onboarding of future GenAI initiatives.

Environment: Python, PySpark, Pandas, NumPy, Scikit-learn, BERT, LangChain, GPT-4o, OpenAI API, AWS Bedrock (Claude, Titan, Cohere), SageMaker, FAISS, VectorDB, RAG (Retrieval-Augmented Generation), Streamlit, Snowflake, Airflow, DBT, Terraform, SQL, Shell Scripting, Amazon S3, AWS Step Functions, Docker, Jenkins, Git, Prompt Engineering, LLM Evaluation, Experimentation Frameworks, Streamlit, Data Modeling, Feature Engineering, REST APIs, CI/CD, Splunk, Google Workspace (Looker/Sheets), Jira, Confluence

Argogroupus, New York, NY

Oct'2021- June'2023

Senior Data Scientist - Fraud & Risk Analytics

- Spearheaded the development and deployment of **ML models** for **fraud detection** and **claims risk scoring**, leveraging **deep learning** and **ensemble methods** to reduce false positives and accelerate high-risk claim identification.
- Architected and productionized **computer vision pipelines** to extract structured data from scanned claim documents using **PyTorch**, **Tesseract OCR**, and **OpenCV** — reducing manual data entry by 60%.
- Modernized **legacy ETL workflows** by implementing scalable **Talend + Snowflake** pipelines, enabling near real-time access to policy and claims data and cutting batch latency by 40%.
- Operationalized **ML workflows** in collaboration with data scientists, designing **feature stores**, **batch inference jobs**, and **version-controlled scoring logic** integrated into decision systems.
- Containerized **model inference services** with **Docker** and deployed on **Kubernetes (EKS)**, backed by **Redis caching** for low-latency, high-throughput scoring across batch and streaming.
- Orchestrated **multi-stage ML pipelines** using **Airflow**, automating **feature generation**, **model scoring**, and **Snowflake exports**, with dynamic dependencies and built-in data validation.
- Proficient in **Python (NumPy, Pandas, Scikit-learn, PyTorch, FastAPI)** and **SQL** for data manipulation, modeling, and deployment; also experienced with **R** for statistical analysis.
- Mentored and led a team of **junior data scientists and engineers**, establishing **reproducible ML practices**, coding standards, and promoting **MLOps principles** across the team.
- Established **CI/CD pipelines** using **Jenkins**, **Git**, and **Terraform**, supporting reliable deployment of both data and ML assets across environments.
- Instrumented robust **model monitoring and alerting** via **Splunk** and custom validation logic, enabling **real-time anomaly detection**, **data drift tracking**, and proactive performance tuning.

- Delivered measurable impact in **financial risk** and **fraud detection**, contributing to operational cost reduction and improved **claims triage** through **ML-driven insights**.

Environment: Python, SQL, Talend, Snowflake, Airflow, Docker, **Kubernetes (EKS)**, Jenkins, Terraform, Git, Amazon S3, Parquet, Avro, JSON, Batch Inference, Feature Stores, CI/CD, Monitoring, JIRA, Agile

Walgreen, Chicago, IL
Senior Machine Learning Engineer

Jan'2020- Sep '2021

- Developed and deployed **machine learning models** to improve **customer targeting**, **pharmacy demand forecasting**, and **personalized healthcare promotions**, supporting retail and clinical use cases across Walgreens' national footprint.
- Designed and maintained **scalable ETL pipelines** using **Azure Databricks**, **Python**, **Apache Spark**, and **Airflow** to process and prepare large-scale **transactional**, **loyalty**, and **prescription** data for analytics and model training.
- Played a key role in Walgreens' **digital transformation** by adopting a **lakehouse architecture** on the **Databricks Data Intelligence Platform** to unify **structured and unstructured data**, enabling **high-speed insights** across **petabyte-scale datasets**.
- Built and evaluated predictive models using **Scikit-learn**, **XGBoost**, and **TensorFlow**, enabling capabilities like **Rx fill abandonment prediction**, **churn detection**, and **next-best-offer generation**.
- Leveraged **Databricks notebooks** and **Delta Lake** for collaborative development, accelerating **experimentation** and **production deployment** of ML solutions.
- Worked closely with analysts, pharmacists, and product managers to translate **business requirements** into deployable ML solutions, aligning with both **clinical** and **retail workflows**.
- Deployed models at scale using **AWS SageMaker** and **Docker**, with **CI/CD automation** via **Jenkins** and **Terraform** for reliable, repeatable production rollouts.
- Implemented **monitoring dashboards** in **Splunk**, including custom logging to detect **feature drift**, **prediction anomalies**, and **pipeline health issues**.
- Collaborated with compliance and infrastructure teams to build **HIPAA-compliant ML workflows**, ensuring secure handling of **sensitive healthcare data** throughout training, inference, and storage.
- Developed internal **Python libraries** and reusable components for **data preprocessing**, **feature generation**, and **batch scoring**, improving consistency and reducing model development time.
- Partnered with engineering teams to integrate **ML predictions** into backend systems and decision flows, improving **prescription adherence**, **campaign efficiency**, and overall **patient engagement**.
- **Mentored junior engineers** on ML modeling, **data pipeline optimization**, and **production readiness** using **Databricks** and **MLOps best practices**.

Environment: Python (NumPy, Pandas, Scikit-learn, PyTorch, FastAPI), SQL, Talend, Snowflake, Amazon S3, Parquet, Avro, JSON, Docker, Kubernetes (EKS), Airflow, Redis, Batch Inference, Feature Stores, Jenkins, Git, Terraform, CI/CD Pipelines, Splunk, Custom Logging, Model Drift Monitoring, JIRA, Agile, Cross-functional ML Ops enablement

Client: Sonata Software, India
Data Scientist

July'17- Sep '19

- Designed and implemented machine learning models using **Python (Scikit-learn, XGBoost, TensorFlow, PyTorch)** to support key business use cases including **fraud detection**, **churn prediction**, and **product adoption scoring**.
- Built real-time data pipelines using **Apache Kafka** and **Spark Streaming** to process and analyze high-velocity data from web logs and user activity for **context-aware ad placement** and **behavioral targeting**.
- Created NLP pipelines with **SpaCy**, **NLTK**, and **TensorFlow-based LSTM models** to extract sentiment, themes, and trends from unstructured customer feedback and social data, enabling brand monitoring and market intelligence.
- Developed and experimented with **custom PyTorch models** (e.g., BiLSTM with attention) for deep text classification and behavioral prediction, improving model flexibility and interpretability.
- Conducted **feature engineering**, model evaluation, and **hyperparameter tuning**, achieving a **30% increase in precision/recall** across predictive pipelines.
- Collaborated with application teams to **productionize models** via **RESTful APIs**, enabling seamless integration with real-time decision systems and recommendation engines.

- Supported **Master Data Management (MDM)** initiatives by aligning and cleansing customer and product data across siloed systems to ensure consistency and accuracy.
- Delivered interactive dashboards and visualizations using **Tableau** and **Power BI**, translating complex analytical results into **executive-ready business insights**.

Environment: Python, SQL, R, Apache Spark (Streaming), Kafka, Scikit-learn, XGBoost, SpaCy, NLTK, Tableau, Power BI, REST APIs, JSON, Git, Jupyter, Shell Scripting, Oracle, PL/SQL

Client: Next Brick Solutions Ltd, India
Data Engineer

Feb, 15- June'17

- Designed and implemented robust **ETL workflows** using Python and SQL to ingest, clean, and transform large volumes of unstructured enterprise content from diverse web portals, CMS platforms, and document repositories.
- Built scalable data pipelines using tools like Talend and Pentaho, enabling high-throughput data processing and integration across Fortune 500 client environments.
- Integrated enterprise search platforms such as **Apache Solr and Elasticsearch**, significantly improving document indexing, search accuracy, and response times for internal knowledge management systems.
- Optimized AWS-based storage and compute infrastructure, including S3 and EC2, to manage cloud-based data processing workloads, improving performance and reducing storage costs.
- Collaborated with cross-functional teams in a hybrid onsite-offshore model to deliver end-to-end data solutions — balancing cost efficiency with technical scalability and rapid delivery timelines.
- Developed and scheduled automated data validation and QA routines, ensuring high levels of data quality, metadata consistency, and schema compliance across incoming datasets.
- Contributed to Java-based backend components using the Spring Framework to support data ingestion, search indexing, and document access pipelines.
- Enabled asynchronous messaging and processing using **RabbitMQ**, facilitating low-latency event-driven data updates and integration between microservices.

Environment: Python, SQL, Shell Scripting, Apache Solr, Elasticsearch, Talend, Pentaho, Oracle 10g, PL/SQL, Java, Spring Framework, Eclipse, Git, JavaScript, jQuery, RabbitMQ, JUnit